

Secure Healthcare Backend with DevSecOps CI/CD Pipeline

Project Overview

This project implements a secure backend for a medicine delivery system using Spring Boot. It demonstrates how DevSecOps integrates development, security, and operations into a unified pipeline.

Application Domain

The system belongs to the healthcare domain, enabling users to register, login, view medicines, and place orders. Admin users can manage medicine inventory.

Technology Stack

Backend: Spring Boot, Spring Security, JWT

CI/CD: Jenkins

Containerization: Docker

Orchestration: Kubernetes

Monitoring: ELK Stack

Security Tools: SonarQube, Trivy

DevSecOps Implementation

Security is integrated at three levels:

1. Application Security: JWT authentication, role-based access, password hashing.
2. Pipeline Security: Secure credentials, static code analysis (SonarQube), container scanning (Trivy).
3. Monitoring Security: Logging of failed logins and unauthorized access using ELK.

CI/CD Pipeline Flow

Code Push → Jenkins Build → SonarQube Scan → Docker Build → Trivy Scan → Push Image → Kubernetes Deployment → ELK Monitoring

Expected Outcome

A fully automated DevSecOps pipeline with secure deployment and real-time monitoring, demonstrating modern software engineering practices.